Incentives for Rural Households to Establish Tree Cover on Agricultural Land in Andhra Pradesh, India

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Abstract

We explore the incentives of farmers to establish forest cover on agricultural land in Andhra Pradesh, India. A large body of theoretical economic work on land-use decisions examines the opportunity costs of switching land uses from agriculture to forest production (Parks and Hardie 1995; Haim et al. 2011; Amacher et al. 2009), but empirical work that estimates the drivers of margin changes for these problems is still needed (Parks et al. 1998). Based on a large dataset collected using a randomized survey of farm households in the East and West Godhavari regions in Andhra Pradesh, we use a multi-level mixed effects probit model to analyze the decision to plant trees on private lands. We find that the most important positive drivers in the decision to plant trees are total land area, years of land tenure, and off-farm labor opportunity. However, factors that lower tree planting incentives are irrigation availability, average travel time to managed plots, elevation, and belonging to the scheduled tribes social group. Moreover, we find evidence that the probability of tree planting is not only associated with important fixed effects (the same across all households in our sample), but it is also associated with random effects that vary from village to village and with the number of plots that the household manages. The decision we study is relevant to public goods production, such as carbon storage or reducing degradation of soil and pressure on native forest cover stocks. The estimates of farmer responses to different relevant variables that we present are readily observable, and can, in principle, be used to support the design of policy by the Indian government to achieve its afforestation and reforestation targets.

1. Introduction

India's forest cover transitions and land-use change have been a focus of large-scale projects carried out by governmental and non-governmental agencies over the past fifty years, most often to ameliorate degraded and previously removed forests (Kohlin et al. 2001; Kohlin and Amacher 2005). The main objectives of these projects have been to improve fuel collection and opportunities for timber use in rural villages through the establishment of community forests (Kohlin et al. 2001; Kohlin and Ostwald 2001). However, a series of policies and other obstacles limiting land holding size and expansion, and the continuing abject poverty among the rural population, have prevented large-scale establishment of forests on private non-community agricultural land (Joshi et al. 2010).

The incentives of farming households in tropical developing countries to establish tree plantations on private land (often called 'trees outside of forests' or TOF) is a relatively understudied topic. There has been some discussion of plantation establishment at a macroeconomic scale in the forest transitions literature, but this is mostly based on modeling reforestation as a replacement land use for tropical forest removal (e.g. see Barbier et al. 2017 for a recent survey). Nonetheless, understanding incentives at the household level to establish forest cover in arid areas, especially where property rights are less than formal, remains critical in predicting the performance of afforestation and reforestation programs worldwide.

The scientific literature demonstrates that secondary forests can generate benefits at the local to global scale by augmenting food and fiber production, reducing harvesting pressure on native forests, and supporting climate mitigation through carbon sequestration (Barbier et al. 2017;

Bastin et al. 2019). Policies that seek to provide financial incentives to establish planted forests through cost sharing of seedlings or information provision programs reflect the perceived value of investing in secondary planting (see Katrin et al. 2019 for an Indonesian example). However, these programs are exceptions and, generally speaking, the decision to plant trees depends critically on location and income constraints, farm and market characteristics, and the economic and demographic status of households. Obstacles to planting forest arise for a host of reasons, not the least of which is the opportunity cost from reduced agricultural income or transaction costs due to a lack of knowledge or experience with tree planting and management. One would expect that transaction costs could be lowered through information programs or social networks that develop in villages and communities; these social fabrics could be critical in predicting reforestation even on regional scales.

In this chapter, we explore the incentives of farmers to establish forest cover on agricultural land in Andhra Pradesh, India. A large body of theoretical economic work on land-use decisions examines the opportunity costs of switching land uses from agriculture to forest production (Parks and Hardie 1995; Haim et al. 2011; Amacher et al. 2009). However, there is scant empirical work that estimates the drivers of margin changes for these problems (Parks et al. 1998). We present analyses based on a large, novel dataset collected using a randomized survey of farm households in the East and West Godhavari regions, which are separated by the Godavari River, the second largest in India, and where a large number of forest processing firms are located. We use a multi-level mixed effects probit methodology to account for unobserved variation across clusters of villages, as well as unobserved variation within villages. The latter is particularly important if the social interactions between farmers within villages might be an

important vehicle to provide information for tree planting. Our estimates serve to identify the key drivers of tree planting decisions on land that can also provide agricultural returns. We present estimates of farmer responses to variables that are readily observable, and which can, in principle, be used to support the design of policy by the Indian government to achieve afforestation and reforestation targets.

2. Household decisions

Our interest in the empirical analysis is to examine the decision to establish tree cover on agricultural land while accounting for unobserved heterogeneity across and within villages. We do not consider specific planting of individual tree species in this chapter; rather, we simply seek to understand why agricultural land is taken out of production, completely or in part, to establish forest cover of any type. The decision we study is relevant to public goods production, such as carbon storage or reducing degradation of soil and pressure on native forest cover stocks.[1]

We present a simple model of decision making by farm households to clarify the choice between planting trees versus maintaining land in an alternative use, namely agricultural production. The decision to plant forest cover depends on the rents from all farm-level decisions as revealed in input use decisions (labor and capital) for the farm as a whole.[2] Given the income and food security links to farm decision making by each household, the decision to plant trees therefore

[1] A plot-based approach, where tree planting is considered on each plot, should be a goal of any future work in this area. Often such approaches are limited by quality and extent of data required for a formal econometric analysis.
[2] For our study area (discussed in the next section), households are often not the owners of formal title for the land they work, but these households still make all production decisions and share income with the owners in a sharecropping system. All households we sample are identical in this regard, and this arrangement is sunk from the perspective of our sampling. As a result, it is valid to assume that the household making crop and tree planting decisions is seeking to maximize profit and thus utility from the farm.

should also depend on the off-farm and market-access opportunities of the household, as well as input use and profitability on the farm.

In what follows, we model decision-making by a representative household *i* in region *j*, but suppress subscripts for brevity (we make the subscripts explicit in the discussion of the econometric specification). A representative household faces the time constraint:

$$T - d = L_c + L_f + L_o \quad (1)$$

where T is total time, d is an (assumed) exogenous measure of travel time to markets and/or field plots,[3] L_c is time the household allocates to crop production, L_f is time allocated to planted tree production, and L_o is time spent by the household in off-farm work. As is conventional in these models, leisure time is not specifically defined but is completely determined by a residual after labor time is determined. Let *L* denote the total time allocated to all income generating activities on the right-hand side of (1), i.e., $L \equiv L_c + L_f + L_o$

Facing the time constraint in (1), the household makes labor and consumption decisions to maximize utility $U(l, C_h; \theta) \equiv U(T - d - L, C_h; \theta)$, which depends on total time allocated to income generating activities *L*, household own-consumption C_h, and a set of exogenous variables that are important to utility, such as household demographics, captured by the vector θ. We make the standard assumptions that $\partial U/\partial C_h > 0$ and that that labor results in disutility (or equivalently, leisure results in utility), i.e., $\partial U/\partial L < 0$.

[3] Quality of roads accessed by the households or distance to the nearest market are examples of a proxy for *d*.

The household also faces a cash constraint requiring total income from all sources to equal total expenditures for farm production and household consumption. Let total household income be given by $I = I_{ex} + I_o + (Q - C_h)P_a + FP_f$, where I_{ex} is exogenous income or assets owned by the household; $I_o = wL_0$ is income earned from off-farm labor work supplied at the exogenous market wage w; Q is household crop production; F is household production of trees; and P_a and P_f are vectors of exogenous market prices for crops and trees, respectively. The term $(Q - C_h)P_a$ represents income earned on the sale of crops produced that are not consumed by the household, if any. When $(Q - C_h)P_a > 0$, the household has positive net production of crops relative to what the household consumes. This surplus is sold at a market price P_a and is part of total income. Conversely, $(Q - C_h)P_a < 0$ implies that the household is a net purchaser of crop goods at the market price.

Using these definitions, the cash constraint requires that total household income equals expenditures:

$$I = I_{ex} + wL_0 + (Q - C_h)P_a + FP_f = wL_H + rK. \quad (2)$$

The right-hand side of equation (2) defines expenditures, including the cost of hired labor, L_H, paid at the market wage rate w, and a vector of all other inputs used in production, denoted K, with unit price vector r.

The variables Q and F in the expression for total household income are household production functions relating outputs to all relevant inputs (labor and nonlabor) as well as other variables that impact production:

$$Q \equiv Q(L_c, K, L_H, \emptyset) \quad (3)$$

$$F \equiv F(L_f, K, \emptyset). \quad (4)$$

In equations (3) and (4), $\emptyset$ is a vector of variables reflecting household experience and farm characteristics important to production. Relevant farm-level variables are similar across production activities in our study region.[4]

We assume that tree planting is a valuable activity to the household through increases in income from selling products derived from trees. This is certainly true in our study area of Andhra Pradesh, because households sell palm oil, fruits and nuts, or timber from species such as casuarina and eucalyptus. In addition, we have not modeled the dynamic nature of tree production for timber species (such as rotation age decisions), because our data can only be used to focus on the one-time decision to plant trees on the farm. As a consequence, we assume that the farm household understands that the rents from trees are a function of rotation decisions, and thus one could re-interpret FP_f in equation (2) as a present value.

The household's problem can now be written as one of choosing consumption, labor allocation, and the use of hired labor and other inputs, to maximize utility subject to the cash constraint:

$$\max_{L, C_h, L_H, K, C_p} U(T - d - L, C_h; \theta) \quad (5)$$

$$s.t.\ I_{ex} + wL_0 + (Q(L_c, K, L_H, \emptyset) - C_h)P_a + F(L_f, K, \emptyset)P_f = wL_H + rK$$

[4] For simplicity, we have not used separate notation for inputs other than labor. This is because, in our sample, most if not all hired labor is used in crop production. We have also not included notation related to plot-level production on the farm. These simplifications do not detract from our main purpose, but would greatly complicate the presentation. Because we focus here only on the decision to plant trees, we will not estimate equations (3) and (4), so it does not matter here whether there is intercropping or single cropping among these outputs, nor does the pattern of input use matter for the farm as a whole. The definition of farm profit here follows from (2) – (4) and is equal to $P_a Q + FP_f - wL_H - rK$.

where we have substituted equations (3)-(4) into equation (2). The solution to this problem is derived by finding the first-order conditions and solving the recursive system of equations first for production decisions, and then for the choice of consumption. First order conditions and analysis of similar problems can be found in more detail in Sills et al. (2003).

3. Welfare and the tree planting decision

For the purposes of the tree planting decision, the problem in (5) is used to define the difference in household welfare from planting trees versus not planting trees. Welfare is defined as the indirect (maximized) utility achievable by the household.[5] The indirect utility function is derived by substituting the optimal choices for consumption, labor times, farm input use, and hired labor into the utility function:

$$V \equiv U(T - d - L^*, C_h^*; I_{ex}, \theta). \quad (6)$$

In equation (6), L^* and C_h^* are optimal labor and consumption decisions, respectively. Recall that total time T and access d are exogenous variables. From the definition of utility and equations (3)-(4), the optimal labor and consumption decisions are functions of prices as well as farm and household characteristics, i.e., $L^* = L(P_a, P_f, r, w, \emptyset, \theta, d)$ and $C_h^* = C(P_a, P_f, r, w, \emptyset, \theta, d, I_{ex})$. The optimal on-farm labor decision L^* is a vector of time allocated to forest and crop production and hired labor; this follows from the first order condition of the farm profit function defined as the total value of production less the total costs, $P_a Q + FP_f - wL_H - rK$. The optimal labor vector also includes off-farm time allocation. Notice in (6) that while exogenous income I_{ex}

[5] For a tree planting example of this welfare measure on a household model in different context, see Amacher et al. (2004).

factors into the consumption decision due to the utility maximization problem of the household, it does not factor into farm profit maximization.[6]

Substituting the expressions for L^* and C_h^* into the utility function, we arrive at a welfare expression rewritten explicitly as a function of all of the variables that are elements of L^* and C_h^* and other exogenous factors of utility and production $(d, \emptyset, \theta)$:

$$V \equiv U(P_a, P_f, r, w, \emptyset, \theta, d, I; F = 0,1). \quad (7)$$

This indirect utility function includes the argument $F = 1,0$, which indicates that there is potentially a different welfare level for households that plant trees ($F = 1$) versus households that choose not to plant trees ($F = 0$). This is true because rents to agricultural production differ depending on whether the household plants trees or not; this results in different optimal choices and, therefore, household welfare. Also, the benefits of planting trees to welfare are not purely financial. Rather, the fact that the utility function drives the tree planting decision for the household implies that non-financial considerations and benefits could be important in the decision for a household to establish forest plantations.

4. Econometric specification of the tree planting decision

The household will plant trees only if doing so results in higher indirect utility than not planting. The decision to plant trees is captured by F_{ij}, which takes on a value of 1 or 0 depending on

[6] This occurs here because the model is assumed separable in production and consumption decisions given that the farm economy in our sample region in Andhra Pradesh is one where labor is supplied and hired freely due to the scale of agriculture common among the villages in the data.

whether household *i* in region *j* plants trees or not, respectively.[7] Observation of F_{ij} depends on unobserved indirect utilities through the following stochastic relationship:

$F_{ij} = 1 \; if \; V_{1ij} + e_1 > V_{0ij} + e_0;$

$F_{ij} = 0 \; otherwise.$ (8)

In equation (8), e_1 and e_0 are errors that may or may not be household specific, as discussed below.

The indirect utilities, V_{1ij} and V_{0ij}, in equation (8) are not observable by the researcher, nor are the error terms e_1 and e_0. Therefore, we define a probability y_{ij}, which is the probability that household *i* located in region *j* will plant trees on their farm (F_{ij} = 1), which is a function of observable variables, where:

$y_{ij} = \boldsymbol{g}_{ij}'\beta + \epsilon_{ij}, \qquad \epsilon_{ij} \sim N(0, \sigma^2)$ (9)

In equation (9), $\boldsymbol{g_{ij}}$ is a vector of *h* independent variables, β is a *h* x 1 vector of corresponding coefficients to be estimated, and ϵ_{ij} is a random error term. The $\boldsymbol{g_{ij}}$ vector is defined more specifically as $\boldsymbol{g_{ij}} = (1, P_{aj}, P_{fj}, r_j, w_j, \emptyset_{ij}, \theta_{ij}, d_{ij}, I_{ij})$ such that equation (9) incorporates all (explanatory) variables in the welfare function (7). For completeness, note that the underlying assumption in equation (9) is $y_{ij} > 0 \; if \; F_{ij} = 1$, and $y_{ij} = 0 \; if \; F_{ij} = 0$.

Given the structure of our data, the standard assumption in (9) may not be sufficient, as there is reason to believe there is correlation between observations that are included in the same cluster (be it villages within a mandal or households within a village). We therefore decompose the error

[7] Location in our model, as described later, may have two geographical or administrative unit levels. These are the village and *mandal* locations (a mandal is equivalent to a county).

in (9) further into a multi-level mixed effects specification (Guo and Zhao 2000; Gibbons and Hedeker 1997).[8] This specification may be viewed as a type of random effects model that generalizes the structure of the error term ϵ_{ij} into three elements (Oshchepkov and Shirokanova, 2020), each measuring a different unobserved effect on the probability of tree planting by household and location:

$$\epsilon_{ij} = u_{0j} + \boldsymbol{h}'_{\boldsymbol{ij}}\eta_i + \xi_{ij} \quad (10)$$

The first term on the right-hand side of equation (10), u_{0j}, captures unobserved factors that vary systematically by location; it therefore defines a location-specific random intercept term (fixed effect). Examples of factors captured by this term include unobserved types of social organizations, and government or private company influence and technical assistance within a village that lead households to make certain decisions. In the second term on the right-hand side of equation (10), the vector $\boldsymbol{h}_{\boldsymbol{ij}}$ is an *m* x 1 subset of the vector $\boldsymbol{g}_{\boldsymbol{ij}}$, where $m \leq h$. The error η_i is a vector of random effects specific to the individual household. The product $\boldsymbol{h}'_{\boldsymbol{ij}}\eta_i$ therefore defines a set of random slope coefficients for a subset of the explanatory variables. The final term in equation (10), ξ_{ij}, captures an idiosyncratic error that is independent of u_{0j} and η_i.

The error structure for our problem accommodates multiple assumptions about correlation of various clusters in the data depending on the (a priori to estimation) assumed composition of $\boldsymbol{h}_{\boldsymbol{ij}}$. Our data for households are collected in individual villages, and several villages are located in

[8] An alternative to specifying (8) is to simply use dummy variables for different villages or regions in the explanatory vector $\boldsymbol{g}_{\boldsymbol{ij}}$, but doing so would implicitly assume that tree planting differences across regions are constant and there is no correlation within a cluster of households in one location. Thus, to assume this a priori risks rendering estimates for β and associated standard errors asymptotically inconsistent and inefficient, because it would be equivalent with assuming that the error is i.i.d.

mandals. Depending on how the subset vector h_{ij} is defined, we can address correlation between households within an individual village in the sample (we will call this the 2-level mixed effects model); or, we can assume that there is correlation both among households in a village and additionally among villages in a mandal (this more general version will be called a 3-level mixed effects model).

The mixed effects models in either 2- or 3-level form improve over standard random effects or fixed effects error structures and are relatively new to economics. These approaches have been very recently argued as relevant in problems such as ours where cluster-based correlation is possible (see Oshchepkov and Shirokanova 2020 for a discussion). The argument for using mixed effects is that cluster correlation means that the variance and covariance of the error term within each cluster might be nonzero. The model in both 2-level and 3-level form can also be estimated by assuming that the structure of the variance-covariance matrix of these correlations does not have a specific a priori form, similar to general heteroskedastic estimators.[9]

For the purposes of our tree planting study, the 2-level mixed effects error can be specified where it is a priori assumed that households within a village have the error ϵ_{ij} in (10) for tree planting decisions that are correlated, but errors across villages within each mandal are independent. The 3-level mixed effects model has an a priori assumption that there is correlation of the ϵ_{ij} between households in the same village, and also across villages in the same mandal. The 3-level specification captures, for example, correlation across villages within East and West

[9] That is, $\mathrm{E}[(\boldsymbol{h}_{ij}'\eta_i)(\boldsymbol{h}_{-ij}'\eta_{-i})] \neq 0, for\ any\ j, and\ i \neq -i \in j$, and thus the definition of *i* and *j* and the subset vector $\boldsymbol{h}_{ij}$ define the type of mixed effects model.

Godavari. Failure to account for these types of correlation when they are present results in biased coefficient estimates and standard errors.

In the econometric results for tree planting of the next section, we consider both 2- and 3-level mixed effects models and formally test which one is more representative of our data. However, it should be pointed out that regardless of model chosen, the mixed effects specification of the error components uses more information than a simple probit model of tree planting, because the latter would rely on assuming errors across households in a village (or across villages in a mandal) are uncorrelated. Using a model that assumes errors are simple and uncorrelated such as (9) would result in tree planting analyses for rural households that are biased, thus also biasing the policy interpretations of the results. Thus, it appears to us at least that the most general approach should be a new standard for estimating the probability of tree planting and identifying drivers of this probability in village-based economies.

An estimable model of tree planting decisions using the underlying error decomposition in equation (10) is of the following form for the probability of tree cover planting:

$$Pr\left(F_{ij}=1\middle|\boldsymbol{g}_{ij},\boldsymbol{u_j},\boldsymbol{h}_{ij}\right)=Pr\left(y_{ij}>0\middle|\boldsymbol{g}_{ij},\boldsymbol{u_j},\boldsymbol{h}_{ij}\right)=\Psi\left(\boldsymbol{g}'_{ij}\beta+u_{0j}+\boldsymbol{h}'_{\boldsymbol{ij}}\eta_i\right). \quad (11)$$

The parameter vector β in equation (11) is estimated using the maximum likelihood procedure described by Guo and Zhao (2000) and Gibbons and Hedeker (1997). Using the estimated coefficient vector, denoted $\hat{\beta}$, the impact of a unit change in any explanatory variable on the probability that the household will plant trees can be obtained from the marginal effect, $\frac{\partial\Psi\left(\boldsymbol{g}'_{ij}\beta+u_{0j}+\boldsymbol{h}'_{\boldsymbol{ij}}\eta_i\right),}{\partial\boldsymbol{g}_{\boldsymbol{ij}}}$. This effect must be calculated differently depending on whether the explanatory variable of interest is discrete or continuous. These marginal effects are critical to

understand the responses of households to changes in relevant explanatory variables such as market access, costs of inputs, and household and farm characteristics.

5. Data

We use a dataset of approximately 2,000 farming households collected across four villages in the district of East Godavari and sixteen villages in the district of West Godavari. Villages were sampled based on extent of tree cover on surrounding farms using two groupings; specifically, we chose the number of sampled households proportional to total land area visible using recent satellite data in tree or plant cover. This procedure was used for two groups: villages with > 60 hectares tree cover and those with < 60 hectares tree cover. This procedure yielded 18 villages from the group with > 60 hectares of tree cover and two from the group with < 60 hectares of tree cover. We excluded villages with <10 hectares of forest cover (many of which had almost no tree planting on land farmed by residents in the village). Table 7 contains a list of villages sampled and their location; Figure 1 shows the geographical distribution and scale of sampled households relative to natural and human landscape features.

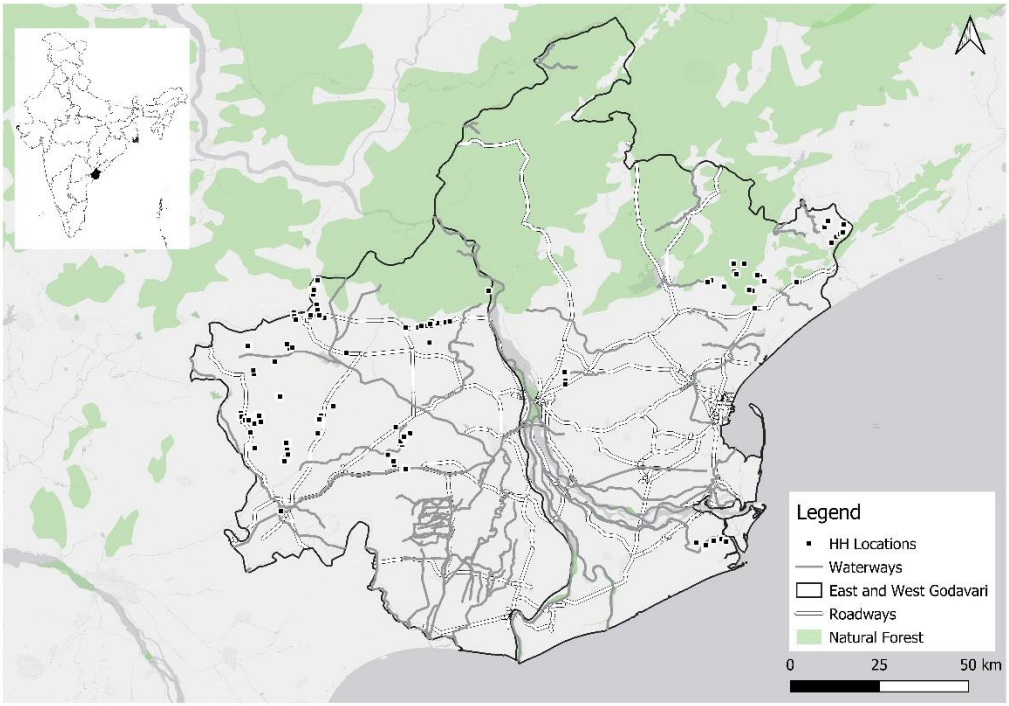


Figure 1. Location of sampled villages in East and West Godavari, Andhra Pradesh

Within each village, we randomly selected households using a simple coin flipping method. In total, we interviewed 2,069 out of 11,747 existing households in the sample villages. Village size varied between 81 households (Lankalapalli, in West Godavari) to 1,980 households (Ungurutu, also in West Godavari). The data were collected between May and July 2019 through a survey instrument containing questions on household demographics, employment and production, community organization and technical assistance, forest management decisions, and plot characteristics, among others.[10] Household production questions (for example: agriculture, grazing, forest management, etc.) were asked for the two main seasons of the year: Kharif (or monsoon season, planted in July and harvested in the months of September and October) and

[10] The survey instrument is available upon request to the authors.

Rabi (planted in November and harvested in the months of May and June). The sampling was preceded by focus group meetings with village leaders and by pretesting individual households in non-sampled villages to refine the survey instrument. In all cases, interviews were conducted with the household member who was indicated as the land-use decision maker, although information in the survey included family-based information as well. Enumerators were hired and trained by the authors from local universities, and a survey management firm in Andhra Pradesh with experience in other university research-based data collection managed coordination of sampling.

Challenges encountered during data collection included an inability to interview a few of the originally selected villages (we were not permitted by the village leaders); this said, all households sampled agreed to be interviewed, answered most of our questions, and indicated whether they planted trees or not. A few households did not provide information on distance to plots for which there was no cultivation during Kharif. Another drawback from the data is that we were not able to gather more detailed information on a plot-by-plot basis due to time constraints. As we discuss below, however, our econometric results are robust. We thus conclude that these limitations have not had a significant impact on our conclusions.

5.1. Descriptive statistics

Tables 1 and 2 present variables used in our analysis and descriptive statistics of the data. In total, 2,062 household heads were interviewed; 592 households (28.6% of the total sample) had planted trees of any species in at least one of their plots, with “tree planting” including tree species planted for various purposes (for example, for fruit or timber production) in family plots

that may be in conjunction with other non-woody species. This way, we can think of our analysis as one of willingness to establish tree cover on farm rather than the choice between specific species (itself an important future research topic).

Table 1 here

In the entire sample, there are on average 1.4 working women, 1.6 working men, and 1.5 children in each household. The average household head is 51 years of age and went through formal schooling for 2.8 years. A total of 1,143 (55%) of interviewed household heads are illiterate. Although discrimination on the basis of castes is illegal in India, the social hierarchical social class system in the country is still present, both in the economic and political spheres, and may for our purposes influence access to education, occupations, income and health resources (Munshi, 2019). In our sample, 16.5% are from “scheduled caste” (SC) groups, 10.5% are from “scheduled tribes” (ST) group, 34.07% are from “other backward caste” (OBC) groups. Among the households that planted trees (N=592), 13.9% of the household heads belong to SC groups, 8.9% belong to ST groups, 39.5% belong to OBC groups, and the remaining 37.8% belong to the FC, suggesting that case could be important to tree planting, although we will formally test this causation in the econometric results.

The average household head had been working the land for 23.4 years, but reported land tenure length in our sample for a household ranges as high as 70 years (most likely a family measure). The average number of plots per household is 1.3 and the average total sum of landholdings per household is 3.7 acres in area (75% of the total sample holds a total of 4 acres or less). Both of

these averages are higher for the group of households that planted trees *versus* households that did not plant trees. The main non-woody species cultivated by households in our sample are maize, rice, cotton, and tobacco. The main tree species planted are cashew (42.9% of households), palm tree for oil production (37.3%) or coconut (10.81%), followed by citrus species (9.9%), and timber and essential oil-producing species (for example, eucalyptus, casuarina, etc.) at 5.6%.

A greater proportion of the group of households that did not plant trees had access to irrigation for crop production (76%) in relation to the group of households that planted trees (60%); considering our total sample, 70% of the households have access to irrigation. The average stated travel time from the households to the closest market (which is how we measured distance in this study) is approximately 25 minutes, whereas the average travel time from the household to one of its plots is 26 minutes. The average household income is Rs.105,593.4 in the year immediately prior to the interview.

Table 2 here

Table 3 presents the main descriptive statistics for the group of interviewed households that planted trees in at least one of their plots and for the group of households that did not plant any trees. Pairwise t-test statistics show that the following variables are statistically different between the two groups: land ownership and tenure length, number of plots managed, travel time to market, sum of travel time to each of the household's plots, number of days working outside of the family's land, and elevation. Although not causative, we find correlation showing that

households that planted trees have also been working their land for a longer period of time and possess a larger number of plots in relation to the households that did not plant trees. Moreover, the sum of the distances to each cultivated plot and that to the closest market are also larger for tree planting households.

Table 3 here

The statistics are reasonable considering that, especially in developing regions, production of food (usually annual crops) is the priority for farmers and it happens on the most fertile and water-abundant lands. Planting trees not only requires higher upfront costs, but it takes years before harvest can be carried out and income generated. Thus, it is common that tree planting will happen on lower quality lands that are located further away from the household and in less irrigated areas. At the same time, tree management requires little labor time after establishment, which is convenient for households with older members or with members working outside of the family's land for part of the year. Finally, we expect social status to influence a household's decision to plant trees, as they should reflect a household's wealth endowment.

6. Econometric Results

Estimation results of the tree planting probability equation (11) are in Table 4.[11] Table 4 shows the estimates for four different specification of the model: Specification 1 and Specification 2 in the first two columns of the table are 3-level mixed effects models that include random effects related to potential error correlation; as such, these specifications assume that errors are

[11] After accounting for missing data, we were left with 1,652 valid observations which are indicated in the results tables.

correlated among villages in a mandal and households in each village.[12] Specifications 3 and 4 in the last two columns assume errors are 2-level mixed effects models that include correlation only among households in each village. Specifications 3 and 4 are also estimated with robust standard errors as they omit the random effect related to the mandal each village is located within. We also include an interaction variable in Specifications 2 and 4 (second and fourth columns of Table 4) to reflect the hypothesis that there is an extra effect on the probability of planting trees for households with larger total land area managed and whose plots are on average more distant.

Referring to the Table, we find statistically significant evidence that the probability that a household will decide to plant trees indeed depends on postulated error components in (10), including exogenous variation accommodated by the fixed effects u_{0j} (which vary by location) as well as the correlation accommodated with the random effects term $\boldsymbol{h}'_{ij}\eta_i$. Tests and goodness of fit measures given in Table 4 show that the mixed effects models fit better than a simple probit one that ignores the error components introduced into (10). The results are similar across the 2-level and 3-level mixed effect estimators, suggesting that the specifications are robust, and that the greatest gains in estimation are from considering correlation in errors across households within a village (rather than villages within a mandal). Percent correct predicted probabilities are also similar and relatively high (greater than 75%) across the four specifications (see Table 6).

[12] Although the theoretical model suggests that input and output prices and costs are an important part of the tree planting decision, in our cross sectional data at one point in time it was the case that these variables had little if any variation given the well-functioning agricultural markets or the unknown price agreements that some farmers undoubtedly have with forest industry and were unwilling to reveal to us in surveying. Two ways our analysis addresses these concerns is first to include exogenous explanatory variable proxies that are measurable and correlated with the opportunity costs and actual costs of tree planting in terms of farm and market access characteristics; and, second, unknown relationships that are price-related between households and forest companies are just one important reason that the mixed effects model is needed, given the unknown correlation that likely exists in villages regarding these.

Table 4 here

Across specifications, variables that are significant and that positively affect the decision of a household in our sample to plant trees are: average travel time from home to plots, total land area managed, whether the household head is the owner of the land, whether the household is engaged in off-farm wage work, household elevation, and the length of land tenure in years. Variables that are significant but negatively affect the decision to plant trees are: whether the household belongs to the "scheduled tribes" social group (in relation to forward caste), whether the household has access to irrigation, whether the household has taken a loan in the past, and average plot area.

Table 5 contains estimated average marginal effects for each variable included in a specification, calculated at the observed levels of the independent variables for each observation, then averaged over all sample observations. A one-year increase in land tenure length is associated with a 2.4% increase in the predicted probability of planting trees, whereas owning the land is associated with a 14.8% higher predicted probability of planting (in relation to not owning the land). Both land tenure length and land ownership are likely a feature of the perception of security with regards to property rights to land in this region. Andhra Pradesh has relatively secure property rights compared to the more frontier rural areas of India where government presence is lower. A secure property rights regime eases the uncertainty of future expropriation, resulting in higher control over the land-use decision cycle and greater likelihood of a longer-term investment such as tree

planting. Long-term decisions may be more attractive if household members know that they will be the ones benefiting from investment.

Table 5 here

Turning to other marginal effects, we estimate on average, that a one-hectare increase in "total land area managed" is associated with a 3.6% increase in the predicted probability that the household will plant trees. However, a one-hectare increase in "average plot area" generates a reduction in the predicted probability of a household planting tree of 3.1%. Despite the apparent contradiction, these opposing effects are reasonable considering that they measure different things. Total land area represents the subsistence and income production potential of the household. Households with larger land area can produce more than enough to sustain the family (food and cash income for essential needs); thus, they have greater possibilities of both reducing agricultural production costs due to economies of scale (if the plots are large enough) and diversifying the number of species cultivated. Assuming that food production and annual income are the two main objectives of the representative household in our region of study, planting trees is considered only after these two objectives are somehow fulfilled. The average size of the plot brings us information on the possibilities of commercial agricultural production for crops and, as mentioned, gains of scale. A household with a larger farmland area has better possibilities or space for tree planting, whereas larger individual plots represent higher potential for production of crops on a commercial scale, making tree planting a less attractive activity.

Elevation and access to irrigation are relevant to our representative household's decision to plant trees, albeit with opposite effects. A one-meter increase in land elevation, which is significant given the topography of our village sample, increases the predicted probability that the household will plant trees in at least one of its lots by approximately 0.4%, while having access to irrigation (a dummy variable) leads to a drop in the predicted probability of 9.4%. Households with access to irrigation systems are likely better off producing annual crops – the land becomes more productive (higher yields) at a relatively low cost (assuming they use flood irrigation methods). However, farmland at relatively higher elevations is usually less productive for crops, and possibly more difficult to irrigate or harder to reach and use machinery. In this case, it might make more sense for the household to plant trees on these lands as trees need less water from irrigation and labor. In the case of a more topographically diverse landscape, tree species that support nontimber products are probably preferable to timber-producing species.

We observe a negative relationship between the probability of planting trees and average travel time to each plot (our measure of distance in this model), though this variable is significant at only the 10% level. For every one-minute increase in the average travel time from the household to a plot, the probability of planting trees decreases by 0.02%.[13] This is counter to our expectation that greater distances from household to the plot favor planting trees versus non-woody crops since it takes longer to reach those plots and tree tending is less labor intensive than crop production. We expected that the greater opportunity cost of time in getting to further plots would be offset by the

[13] This is exogenous for households in our sample, as there is very little choice involved in traveling to plots along the main and accepted paths leaving each village

household through tree planting. However, we also find travel distance to the closest market not to be statistically significant in our model for tree planting.

There is another possibility for the unexpected distance result. We do not have data on unobserved forest industry influences in this region (in part, this is a possible reason for correlation we allow in the model estimation). Indeed, given that most of the tree species planted by interviewed households are cashew and palm and that Andhra Pradesh presents strong and well-organized markets for these products with firms located along the Godavari River, the processing industry might be absorbing some of the production costs of these forest products for households in our study region. Further, households in our sample are likely either taking their production directly to the companies or (most commonly) to intermediaries who purchase at the farm gate rather than traveling periodically to nearby markets.

As far as caste effects are concerned, households belonging to the "scheduled tribes" social group have a 10.6% lower probability of planting trees in relation to the base social group "forward caste." Scheduled tribes households likely have lower exogenous income endowments and heavier reliance on farm only production for subsistence, relative to the forward caste (so that $Q(L_c, K, L_H, \emptyset) - C_h = 0$ in equation 2). Planting trees as an economic activity usually requires high initial planting and opportunity costs (given the longer time that it takes for the investment to be recovered relative to annual crops), especially if the household has a small farmland area to work on and needs to produce for their own subsistence. Moreover, their ability to find credit and perhaps access to formal forest markets is lower, due to the lack of collateral and contacts.

For households in our sample that have members working outside of their own farm, the probability that they will plant trees increases by 0.08% per day of off-farm work. Off-farm work is known to be a strategy to smooth out income and reduce risk exposure to extreme weather events and pests (Lamb, 2003; Ito and Kurosaki, 2009). If the household has members working off the farm, they will prefer to plant and cultivate species that need less attention and labor.

Finally, we should mention that we did not include input prices (for example, fertilizer, herbicides, etc.) in the model. There is very little variation in the data for these variables, which is expected given that output and input markets are complete in the region of study (see Lamb, 2003 for published evidence of this). The implication to our study is that, as a consequence, all households face the same market prices. [14]

7. Discussion and conclusions

We investigate factors determining rural household choices to plant trees on family farms in the East Godavari and West Godavari districts of Andhra Pradesh, India. We use a random utility theoretical model to develop and estimate a multilevel mixed-effects probit model using a maximum likelihood procedure applied to a large survey of randomly sampled households in the region. The mixed effects probit model approach is relatively new in the economics of rural households, but it has the advantage of allowing for random effects correlations among errors in

[14] Indeed, we found that because all households faced the same regional prices, there was very little if any variation in this variable in the sample when reported. We also included herd size (number of buffaloes and cows owned) as an exogenous variable because they may be considered as a sunk asset insofar as these animals are raised to produce milk, dung, and are used for animal traction purposes in agriculture (for evidence, see Bairwa et al., 2013).

the estimation of the tree planting decision at a locational scale, such as among households within villages, or among villages at a regional level. We a priori expect these correlations due to unobserved variables such as transaction costs or forestry company influences among households in the region.

We find evidence that the probability of tree planting is not only associated with important fixed effects, which are the same across all households in our sample, but it is also associated with random effects that vary from village to village and with the number of plots that the household manages. We also find that the most important positive drivers in the decision to plant trees are total land area, years of land tenure, and off-farm labor opportunity. However, factors that lower tree planting incentives are irrigation availability, average travel time to managed plots, elevation, and belonging to the scheduled tribes social group.

There are important forest policy and development implications following from these results. The establishment of secondary forest and tree cover has recently been a focused strategy of the Indian government to reduce pressure over native forests and foster rural development. Moreover, with restrictive policies on native forest harvesting and the low availability of raw material facing the forest products industry in Andhra Pradesh (Schons et al. 2020), companies have started to follow a contract farming model and are increasingly establishing agreements with private farmers to grow trees, in return for forest industry sharing in seedling cost and availability as well as providing technical assistance. Our results are suggestive of the types of households and specific characteristics of the family farm that will provide greatest traction for these agreements and meet the objective of protecting native forests.

As opposed to other regions in India and many developing countries, land tenure and ownership are relatively well defined and enforced in Andhra Pradesh, although there are still questions about the equality of land distribution and traditional rights and access of certain social groups to land. Well defined and enforced rights to land are known to be relevant factors for long-term investment by households in their land. Tree planting constitutes such an investment and, as a direct consequence, so are commitments to carbon sequestration and avoided emissions programs. The fact that our econometric model establishes that length of land tenure and land ownership constitute significant and positive factors for the tree planting decision represents a first step to successful income-generating tree planting farm contracting schemes and REDD+ programs (these are especially relevant for forest-dependent farmers). Such steps are already envisioned by the Indian government in a recent revision of the National Forest Policy (DNFP-2018), but our work provides evidence that this strategy is likely to be successful. In fact, based on the responses we estimate for tree planting among these households, the results suggest that this program could be transformative, at least for Andhra Pradesh.

We also show that greater total landholding size increases the probability that farmer households will plant trees. From the perspective that priority is given to the cultivation of cash and food producing crops in a country as poor as India, tree planting and forest establishment would only be undertaken after subsistence needs are met using the most productive, i.e., higher quality, land. However, considering that larger landholdings are often owned by groups with higher social status, perhaps using carbon sequestration as a complementary incentive mechanism for private planting could be a way of reducing social inequities or improving income possibilities so

that subsistence-based farming is no longer a necessity. Cash transfer programs could additionally be associated with tree planting in tribal private farmlands, perhaps through a REDD+ mechanism that includes native forest enrichment or restoration activities and encourages contracting between rural households and forest products companies. Forest products companies would not only benefit from a more available supply of timber, but households would also be able to increase the value of their final product (maybe through a price premium, but probably through greater market access) with forest management or fair-trade types of certification.

Land quality is also the reason why households located in higher (possibly hilly) grounds present a higher probability of planting trees in our data; households with access to irrigated lands present a lower probability of engaging in the activity. Tree planting has been seen with some skepticism in drought-prone regions, especially because species such as eucalyptus are popularly known for "drinking" all the water from the land (Tripathi and Mishra 2017). Establishment of efficient water irrigation systems that can accommodate both crop and tree production could play an important role in making widescale tree planting a reality.

Our results also seem to show scope for "public-private participation models" in India as proposed in Indian policy documents surrounding DNFP-2018, at least for the case of Andhra Pradesh. Although the analysis in this chapter constitutes a first approximation of the kind to incentives for planting "trees outside of forests" in Andhra Pradesh, further research would contribute significantly to the design of the new policy. This policy is still in the development phase and is intended to support India's responsibilities under the Paris Climate Agreement at the

same time that it builds upon the current Joint Forest Management program (enacted in 1988) and the Forests Rights Act of 2006. The latter aims to give tribal communities greater traditional rights to their lands. If tree planting is to happen on a greater scale, these communities should be given greater ownership of land or management rights.

On a final note, we have discussed determinants of tree planting based on preferences that farmers have revealed to us through household surveys. Our results cannot and should not be used to debate the merits of DNFP-2018. This said, our results do hint at the fact that any long-term forest policy should consider all stakeholders, incentives and laws in unison. Doing so would increase the likelihood that the economic, development and conservation aspects related to forest management efforts that are sustainable over time.

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Table 1 Definition and labels of variables used in the econometric specifications

	Variable name	Definition
Household characteristics	Plant_tree	Households planting trees (1=Yes; 0=No)
	Schooling	Schooling of household head (n. of years)
	Tenure	Length of land tenure
	Off-farm	N. of off-farm workdays
	Herd	Herd size (n. of buffaloes and cows)
	Age	Age of household head (years)
	Coop	Participation in co-operatives (1=Yes; 0=No)
	S_group	1= Scheduled caste (SC)
		2 = Scheduled tribe (ST)
		3 = Other backward caste (OBC)
		4 = Forward caste/other (FC)
Farm characteristics	Loan	Household head has taken loan in the past year (1=Yes; 0=No)
	Plot_avarea	Average plot area (acres)
	Land area	Total worked land area (acres)
	Elev	Elevation (meters)
	Plot_number	Number of plots managed
	Neighbor	N. of neighbors have planted tree
	Irrig	Irrigation availability (1=Yes; 0=No)
	Own_land	Household own the land (1=Yes; 0=No)
Market characteristics	Tr_market	Travel time from home to market (minutes)
	Tr_avplot	Average travel time from home to plot (minutes)
	Tr_allplots	Total travel time for all plots (minutes)

Table 2 Descriptive statistics for continuous variables (first part) and dummy variables (second part)

	Variable name	mean	Standard deviation	min	max
Household characteristics (continuous)	Age	51.11	11.79	22	86
	Schooling	2.78	3.81	0	17
	Tenure	24.27	13.26	1	70
	Off-farm	31.74	43.73	0	350
	Herd	1.38	2.66	0	53
Farm Characteristics (continuous)	Plot_avarea	3.03	3.39	0	70
	Land area	3.72	4.60	0	71
	Elev	38.59	57.73	0	1106
	Plot_number	1.27	0.62	1	7
	Neighbor	0.53	1.56	0	20
Market characteristics (continuous)	Tr_market	37.12	23.81	1	230
	Tr_avplot	25.86	16.28	0	120
	Tr_allplots	30.89	20.93	0	250

	Variable name	1=Yes	0=No
Household characteristics (dummy)	Plant_tree	592 (28.62%)	1477 (71.39%)
	Coop	197 (9.52%)	1872 (90.49%)

	Variable name	1=Yes	0=No
Farm characteristics (dummy)	Irrig	1179 (70.35%)	497 (29.65%)
	Own_land	1908 (92.22%)	161 (7.78%)

Table 3 Pairwise differences and t-test statistics significance tests for households that plant trees versus household that do not plant[1]

Group of variables	Variable name	Households that do not plant trees	Households that plant trees	Difference (Std. error)
Household characteristics	Schooling	2.900	2.553	0.348 (0.198)
	Tenure	24.05	24.70	-0.649 (0.689)
	Off-farm	28.91	37.26	-8.343*** (2.265)
	Herd	1.427	1.272	0.155 (0.138)
	Age	51.13	51.04	0.0861 (0.613)
	Coop	0.104	0.0841	0.0202 (0.0154)
	S_group	2.896	3.011	-0.115* (0.0554)
	SCs	0.177	0.136	0.0415* (0.0192)
	STs	0.125	0.0894	0.0359* (0.0165)
	OBCs	0.321	0.403	-0.0814** (0.0247)
	FCs	0.376	0.372	0.0039 (0.0252)
Farm characteristics	Loan	0.0961	0.0590	0.0370* (0.0144)
	Plot_avarea	3.106	2.874	0.232 (0.176)
	Land area	3.511	4.129	-0.618** (0.239)
	Elev	37.08	41.55	-4.470 (3.000)
	Plot_number	1.191	1.433	-0.242*** (0.0317)
	Neighbor	0.555	0.490	0.0652 (0.0811)
	Irrig	0.755	0.597	0.157*** (0.0235)
	Own_land	0.909	0.980	-0.0709*** (0.0129)
Market characteristics	Tr_market	35.75	39.78	-4.026** (1.235)
	Tr_avplot	25.78	26.02	-0.245 (0.847)
	Tr_allplots	28.96	34.69	-5.730*** (1.079)

1. Standard errors in parentheses; significant levels are *** p<0.01, ** p<0.05, * p<0.1

Table 4 Multilevel mixed effects probit model estimation results for a priori assumed 2- and 3-level correlation[1]

Explanatory variables	Specification 1 3-Level mixed effects without interaction variables	Specification 2 3-Level mixed effects with interaction variables	Specification 3 2-Level mixed effects without interaction variables and robust standard errors	Specification 4 2-Level mixed effects with interaction variables and robust standard errors
Schooling	0.0052	0.0044	0.0053	0.0045
	(0.0113)	(0.0113)	(0.0161)	(0.0163)
Tenure	0.0107***	0.0108***	0.0108***	0.0109***
	(0.0032)	(0.0032)	(0.0025)	(0.0025)
Off-farm	0.00349***	0.00358***	0.00348***	0.00357***
	(0.0009)	(0.0009)	(0.0013)	(0.0013)
Herd	0.0044	0.0043	0.0043	0.0048
	(0.0155)	(0.0155)	(0.0159)	(0.0160)
Age	-0.0035	-0.0035	-0.0036	-0.0036
	(0.0037)	(0.0037)	(0.0033)	(0.0033)
Coop	-0.161	-0.151	-0.163	-0.153
	(0.135)	(0.135)	(0.163)	(0.162)
SC	-0.149	-0.154	-0.146	-0.151
	(0.123)	(0.123)	(0.116)	(0.115)
ST	-0.511***	-0.526***	-0.510**	-0.525**
	(0.146)	(0.146)	(0.230)	(0.226)
OBC	0.165*	0.166*	0.168	0.169
	(0.0957)	(0.0958)	(0.117)	(0.117)
Loan	-0.214	-0.228	-0.211	-0.224
	(0.154)	(0.154)	(0.179)	(0.178)
Plot_avarea	-0.115***	-0.134***	-0.118*	-0.138**
	(0.0429)	(0.0451)	(0.0717)	(0.0692)
Land_area	0.143***	0.131***	0.147**	0.134*
	(0.0391)	(0.0398)	(0.0687)	(0.0714)
Elev	0.00191**	0.00189**	0.00188***	0.00186***
	(0.0007)	(0.0007)	(0.0005)	(0.0006)
Plot_number	0.174	0.171	0.157	0.154
	(0.158)	(0.157)	(0.179)	(0.180)
Neighbor	0.0142	0.0130	0.0141	0.0129
	(0.0261)	(0.0261)	(0.0224)	(0.0227)
Irrig	-0.431***	-0.423***	-0.434**	-0.426**
	(0.107)	(0.107)	(0.192)	(0.193)
Own_land	0.776***	0.769***	0.778***	0.771***
	(0.222)	(0.222)	(0.238)	(0.243)
Tr_market	-0.0016	-0.0018	-0.0015	-0.0018
	(0.0019)	(0.0019)	(0.0029)	(0.003)
Tr_avplot	-0.0106	-0.0130*	-0.0107	-0.0131*
	(0.0075)	(0.0077)	(0.0071)	(0.0076)
Tr_allplots	0.0060	0.0048	0.0061	0.0049
	(0.0061)	(0.0062)	(0.0054)	(0.0054)
	(0.0532)	(0.0513)		
variance (villages)[2]	0.521	0.523		
	(0.344)	(0.344)		
covariance (villages)	0.0311	0.0340		
	(0.0965)	(0.0949)		
(Land_area) x (Tr_avplot)		0.00118*		0.0012

		(0.0007)		(0.0009)
variance (households)[3]			0.0595	0.0566
			(0.0423)	(0.0405)
covariance (households)			0.890**	0.897**
			(0.420)	(0.422)
Constant	-1.690***	-1.581***	-1.661***	-1.553***
	(0.451)	(0.456)	(0.395)	(0.409)
Observations	1,652	1,652	1,652	1,652
Number of groups	16	16	22	22
Log likelihood	-695.204	-693.747	-695.904	-694.460
Wald chi	122.53	125.35	2537.91	7206.63
Prob>chi2	0.000	0.000	0.000	0.000
AIC	1440.408	1439.495	1433.81	1430.92

1. Standard errors in parentheses; significance indicated at following levels: *** p<0.01, ** p<0.05, * p<0.1
2. Significance indicates 2-level mixed effects with error correlation among villages within a mandal
3. Significance at household level indicates 2-level mixed effects

Table 5. Estimated marginal Effects by model specification[1]

Explanatory variable[2]	**Specification 1** 3-Level mixed effects without interaction variables	**Specification 2** 3-Level mixed effects with interaction variables	**Specification 3** 2-Level mixed effects without interaction variables and robust standard errors	**Specification 4** 2-Level mixed effects with interaction variables and robust standard errors
Schooling	0.0012	0.0010	0.0012	0.0011
	(0.0027)	(0.0027)	(0.0038)	(0.0039)
Tenure	0.0025***	0.0026***	0.0026***	0.0026***
	(0.0008)	(0.0008)	(0.0006)	(0.0006)
Off-farm	0.0008***	0.0008***	0.0008***	0.0008***
	(0.0002)	(0.0002)	(0.0003)	(0.0003)
Herd	0.001	0.0011	0.001	0.0011
	(0.0037)	(0.0037)	(0.0037)	(0.0038)
Age	-0.0008	-0.0008	-0.0008	-0.0008
	(0.0009)	(0.0009)	(0.0008)	(0.0008)
Coop	-0.0375	-0.0351	-0.0378	-0.0354
	(0.0309)	(0.0311)	(0.0372)	(0.0371)
SC	-0.0349	-0.0361	-0.0339	-0.0350
	(0.0286)	(0.0286)	(0.0268)	(0.0267)
ST	-0.110***	-0.113***	-0.110**	-0.113**
	(0.0327)	(0.0328)	(0.0491)	(0.0484)
OBC	0.0408*	0.0411*	0.0413	0.0416
	(0.0241)	(0.0241)	(0.0291)	(0.0291)
Loan	-0.0494	-0.0523	-0.0485	-0.0514
	(0.0346)	(0.0347)	(0.0403)	(0.0399)
Plot_avarea	-0.0273**	-0.0319***	-0.0281	-0.0326*
	(0.0107)	(0.0114)	(0.0172)	(0.0167)
Land_area	0.0341***	0.0384***	0.0348**	0.0390**
	(0.0101)	(0.0107)	(0.0167)	(0.0166)
Elev	0.0004**	0.0004**	0.0004***	0.0004***
	(0.0002)	(0.0002)	(0.000)	(0.0001)
Plot_number	0.0499	0.0491	0.0493	0.0484
	(0.0397)	(0.0395)	(0.0417)	(0.0418)
Neighbor	0.0034	0.0031	0.0033	0.003
	(0.0062)	(0.0062)	(0.0053)	(0.0054)
Irrig	-0.106***	-0.104***	-0.106**	-0.104**
	(0.0292)	(0.0291)	(0.0482)	(0.0482)
Own_land	0.158***	0.157***	0.159***	0.158***
	(0.0427)	(0.0428)	(0.0451)	(0.0460)
Tr_market	-0.0004	-0.0004	-0.0004	-0.0004
	(0.0004)	(0.0004)	(0.0007)	(0.0007)
Tr_avplot	-0.0025	-0.002	-0.0025	-0.0021
	(0.0018)	(0.0018)	(0.0017)	(0.0017)
Tr_allplots	0.0014	0.0011	0.0014	0.0011
	(0.0015)	(0.0015)	(0.0013)	(0.0013)
Observations	1,652	1,652	1,652	1,652

1. Standard errors in parentheses; All marginal effects are calculated at mean values across observations
2. significance levels are as follows: *** p<0.01, ** p<0.05, * p<0.1

Table 6. Percent correct predictions of tree planting provability

Model prediction		**Specification 1** 3-Level mixed effects without interaction variables			**Specification 2** 3-Level mixed effects with interaction variables			**Specification 3** 2-Level mixed effects without interaction variables and robust standard errors			**Specification 4** 2-Level mixed effects with interaction variables and robust standard errors		
		Household[1] planted trees		Number observations	Household planted trees		Number observations	Household planted trees		Number observations	Household planted trees		Number observations
		0=No	1=Yes		0=No	1=Yes		0=No	1=Yes		0=No	1=Yes	
0=No	Frequency	972	179	1,151	975	176	1,151	974	177	1,151	977	176	1,153
	Percentage predicted (%)	**84.45**	15.55	100	**84.71**	15.29	100	**84.62**	15.38	100	**84.74**	15.26	100
1=Yes	Frequency	121	380	501	118	383	501	119	382	501	116	383	499
	Percentage predicted (%)	24.15	**75.85**	100	23.55	**76.45**	100	23.75	**76.25**	100	23.25	**76.75**	100

1. stated planting of trees in household survey

Table 7. List of Villages and Mandals covered in survey sample[1]

District	Codes	Village	Codes	Mandal	Codes
West Godavari	1	Rajanagaram	1	Buttayagudem	1
West Godavari	1	Barrinkalapadu	2	Jeelugu Milli	2
West Godavari	1	Lankalapalle	3	Jeelugu Milli	2
West Godavari	1	Parimpudi Narayanapuram	4	Jeelugu Milli	2
West Godavari	1	Vadlapatlanuthanam	5	Kamavarapukota	3
West Godavari	1	Kothapalle	6	Dwaraka Tirumala	4
West Godavari	1	Malleswaram	7	Lingapalem	5
West Godavari	1	Avapadu	8	Nallajerla	6
West Godavari	1	Nyayampalle	9	Pedavegi	7
West Godavari	1	Ramasingavaram	10	Pedavegi	7
West Godavari	1	Bandivarigudem	11	T.Narasapuram	8
West Godavari	1	Tedlam	12	T.Narasapuram	8
West Godavari	1	Gopinadhapatnam	13	Ungutur	9
West Godavari	1	Unguturu	14	Ungutur	9
East Godavari	2	Billananduru	15	Kotananduru	10
East Godavari	2	Doparthi	16	Prathipadu	11
East Godavari	2	U. Jagannadhapuram	17	Prathipadu	11
East Godavari	2	Palacharla	18	Rajanagaram	12
East Godavari	2	Dondavaka	19	Tuni	13
East Godavari	2	Brahmasamedyam	20	Katrenikona	14
West Godavari	1	Erraguntapalle	22	Chintalapudi	15
West Godavari	1	Ayyaparajugudem	23	Lingapalem	16

1. A mandal in India is similar to a county in the U.S.